

AI Resume Analyzer

Intelligent Role Validation and Career Recommendation System

1. Introduction

Recruitment processes in modern organizations are increasingly dependent on automated shortlisting systems. While this improves efficiency for recruiters, candidates often receive no clear indication of whether their resume aligns with the role they apply for. As a result, many applications fail due to skill mismatch and weak targeting rather than lack of potential.

The AI Resume Analyzer addresses this problem by introducing a structured, explainable resume evaluation workflow. It combines document processing, natural language processing (NLP), skill normalization, and weighted role matching to provide pre-application intelligence to job seekers and students.

The system serves as a practical career guidance assistant by showing measurable compatibility, identifying skill gaps, and recommending learning actions. This improves application quality and helps candidates prepare strategically for role-specific opportunities.

2. Problem Statement

Candidates frequently apply to roles without understanding compatibility between their resume and job expectations. Recruiters use ATS-based filtering pipelines that remove low-fit resumes quickly, often before manual review. Students and fresh graduates also face uncertainty in selecting suitable role tracks based on their current capabilities.

Additionally, systems that process uploaded documents without validating document type can generate incorrect analysis for non-resume files such as course notes, assignments, or certificates. A robust, reliable analyzer must detect and reject such files before scoring.

3. Objectives of the Project

- Analyze resumes using NLP-driven text understanding.
- Extract technical skills automatically from document content.
- Compare extracted skills against role-specific requirements.

- Compute compatibility percentages using weighted scoring.
- Identify matched and missing skills for each role.
- Recommend suitable roles and guided learning pathways.

4. System Overview

The system begins with secure upload of a PDF or DOCX file. The backend extracts readable text and runs resume validation checks. If valid, NLP preprocessing and skill extraction modules detect canonical skill entities. These entities are matched against weighted role skill graphs, and a score engine produces role-fit metrics and recommendations.

In role-validation mode, users select a target role and receive a focused compatibility report. In role-discovery mode, the system evaluates all supported roles and returns ranked compatibility results. The frontend dashboard visualizes outcomes through structured cards and charts.

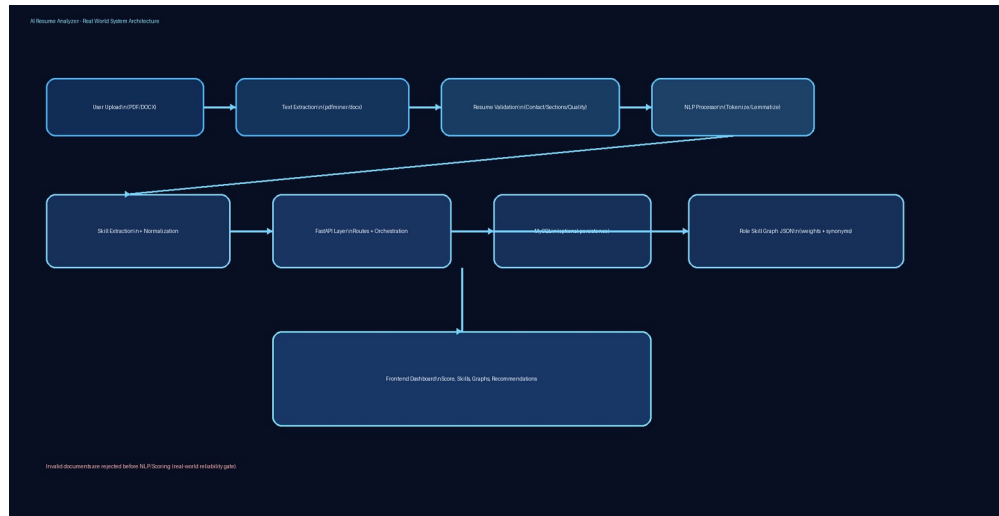
5. System Architecture

Architecture Components:

- User Interface (React frontend)
- Backend API Layer (FastAPI)
- Resume Parser (PDF/DOCX extraction)
- Resume Validation Engine
- NLP Processor and Skill Normalizer
- Role Matching and Weighted Scoring Engine
- Dashboard Visualization Layer

Data Flow: User Upload -> Text Extraction -> Validation -> NLP Processing -> Skill Normalization -> Role Graph Matching -> Scoring -> Dashboard Results.

Figure A1: Real-World Deployment Architecture



The architecture includes strict validation gates before skill scoring and supports both role-graph intelligence and optional relational persistence.

6. Technology Stack

Frontend Technologies: HTML, CSS, JavaScript, React, Tailwind CSS, Framer Motion, Recharts.

Backend Technologies: Python, FastAPI, SQLAlchemy, MySQL.

Libraries Used: pdfminer.six, python-docx, NLTK.

Visualization: Role compatibility bar chart, skill match pie chart, and strength radar chart for interpretability.

7. Natural Language Processing Implementation

1. Resume text extraction from PDF/DOCX inputs.
2. Text cleaning and normalization.
3. Tokenization and stopword removal.
4. Lemmatization for base-form terms.
5. Skill detection from normalized token sets.
6. Synonym normalization into canonical skills.
7. Role graph matching for compatibility evaluation.

This NLP pipeline enables robust matching beyond exact keyword search by handling variants and contextual skill expressions.

8. Resume Validation System

Before scoring, uploaded documents pass through a validation layer that checks contact presence, resume section patterns, minimum readable content, and non-resume indicators. Invalid files are blocked with a clear warning message. This protects system reliability and prevents misleading outputs.

9. Role Matching and Scoring System

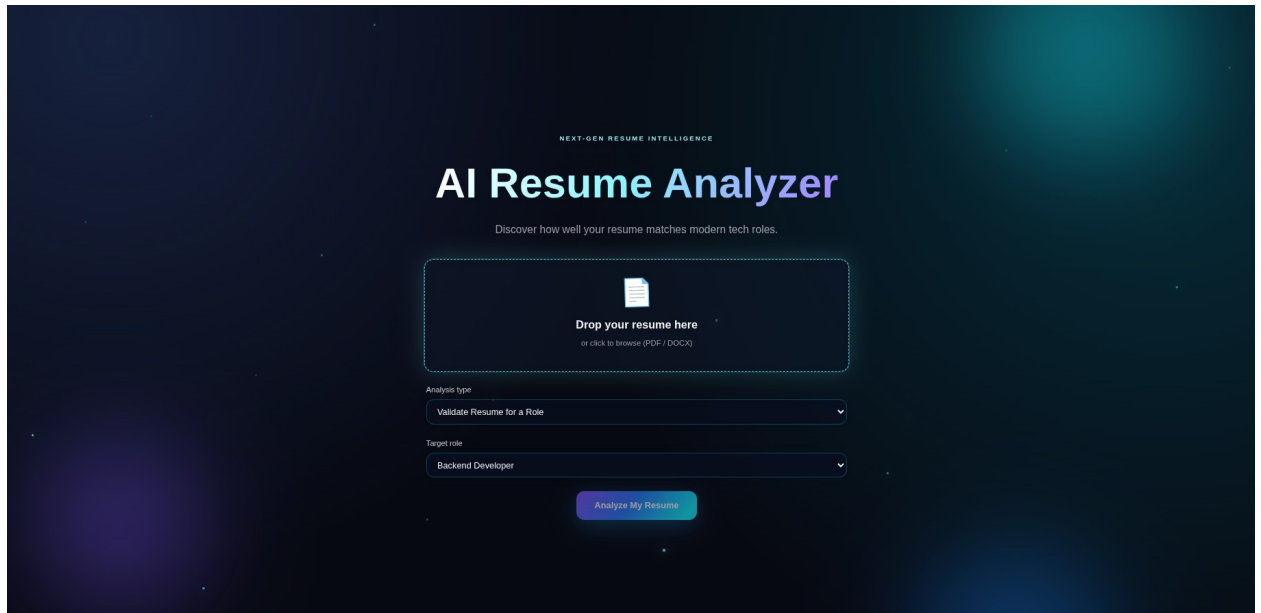
Each role is modeled as weighted skill groups such as Core Skills, Frameworks, Databases, and Tools. The engine computes group-wise coverage and final compatibility percentage. Missing high-impact skills are prioritized to generate recommendations, learning paths, and suggested courses.

10. Key Features

- Resume Upload and Processing
- Automatic Skill Extraction
- Resume Validation System
- Role Compatibility Analysis
- Missing Skill Detection
- Career Role Recommendation
- Interactive Dashboard Visualization

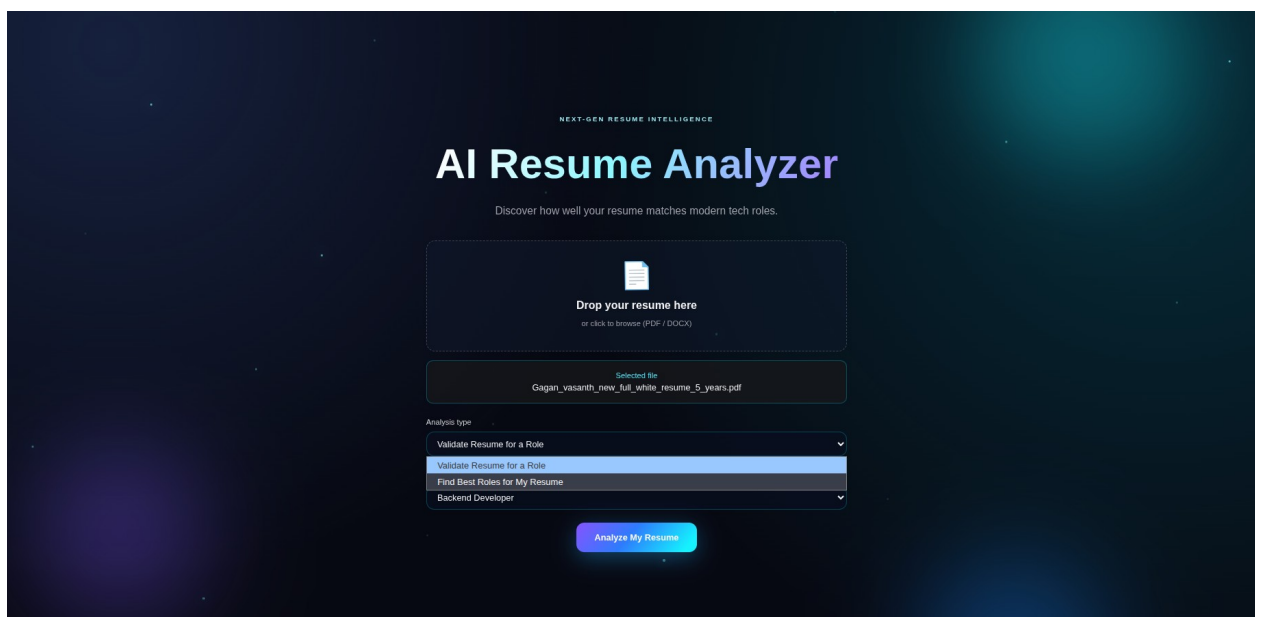
11. Application Screenshots

Figure 1: Landing Page - Resume Upload Interface



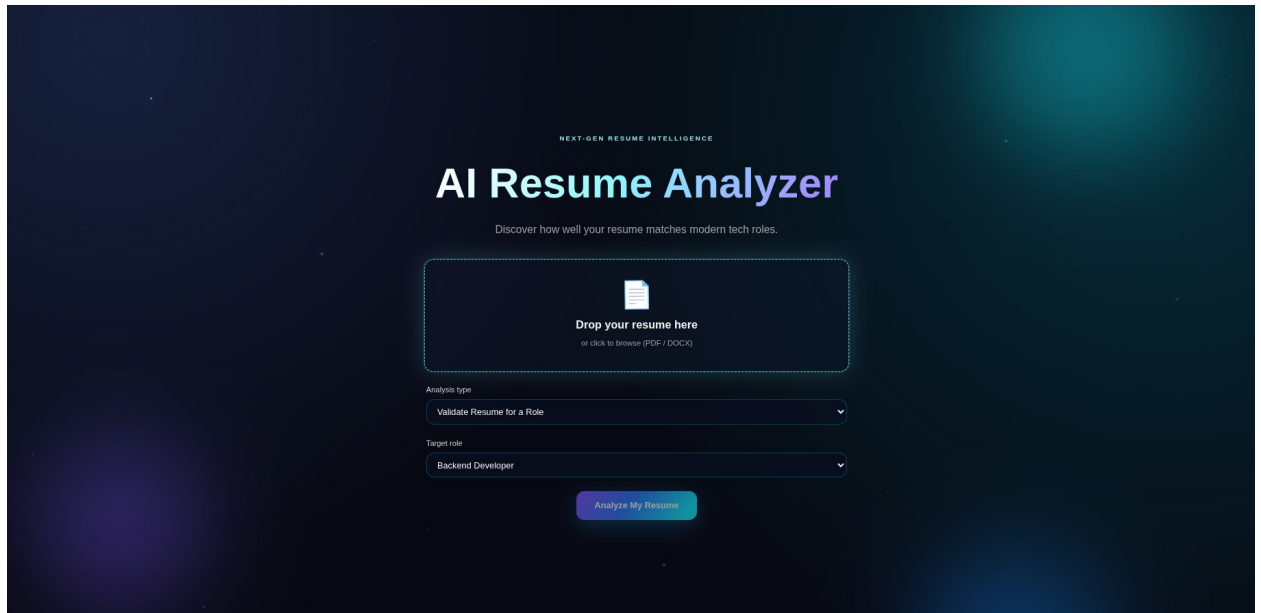
Main landing screen with drag-and-drop upload, analysis mode selection, and role input controls.

Figure 2: Resume Selection Workflow



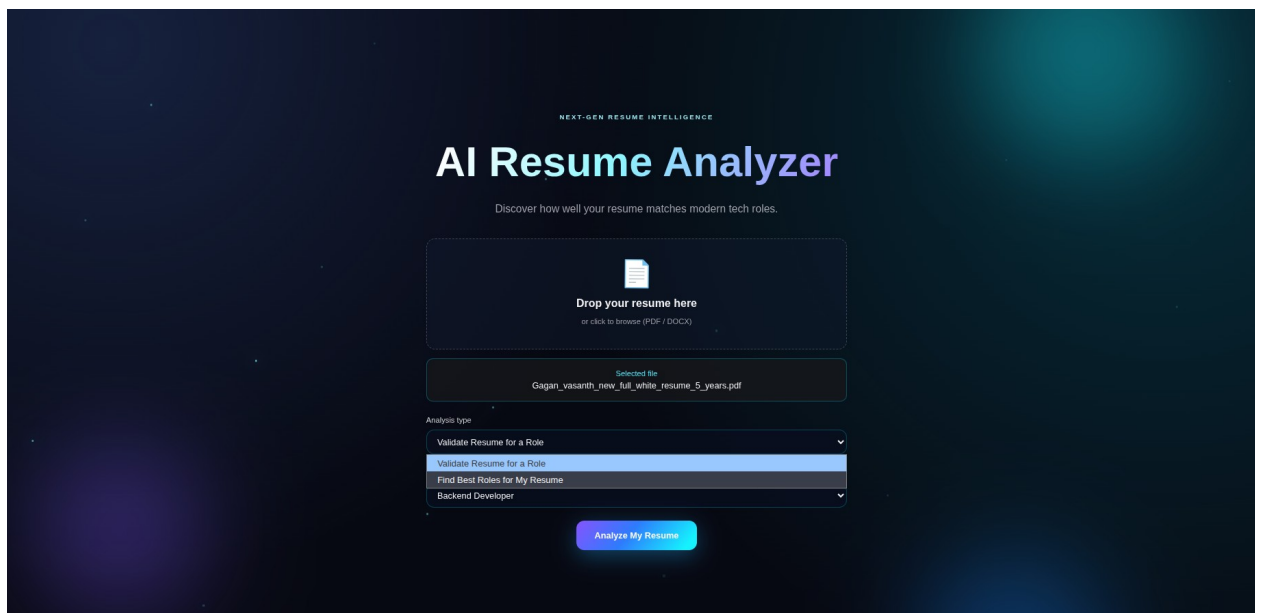
Shows selected resume file preview before analysis submission.

Figure 3: Role Selection Dropdown



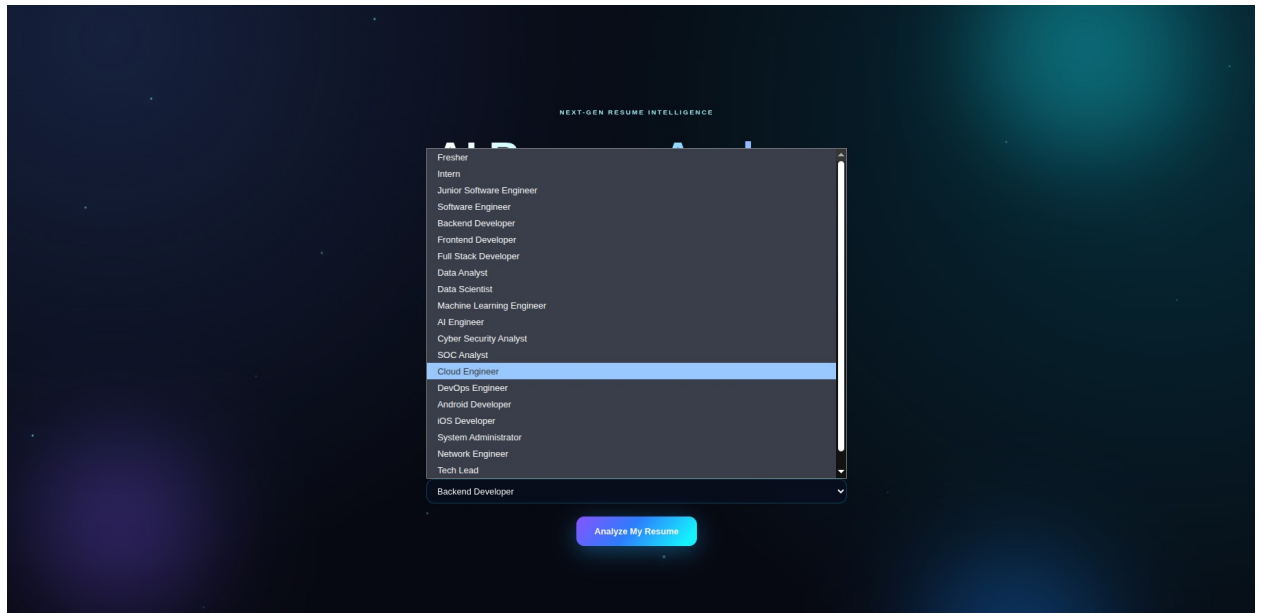
Displays supported career roles for targeted compatibility analysis.

Figure 4: Processing Screen - Stage A



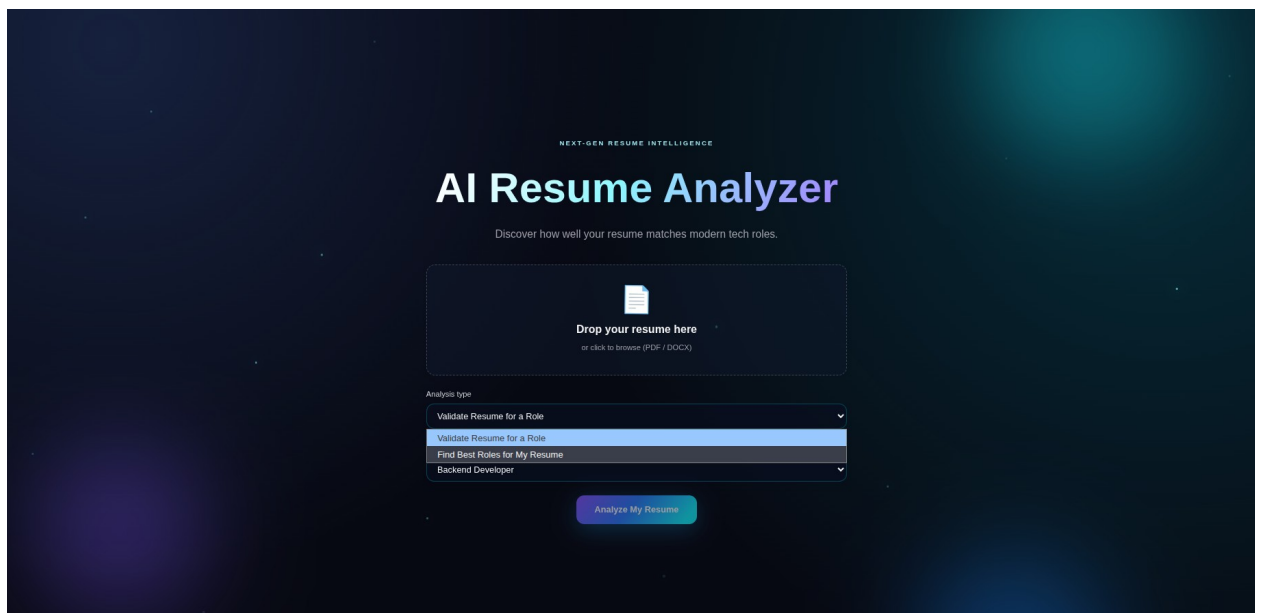
Illustrates analysis progress across upload, extraction, matching, and scoring stages.

Figure 5: Processing Screen - Stage B



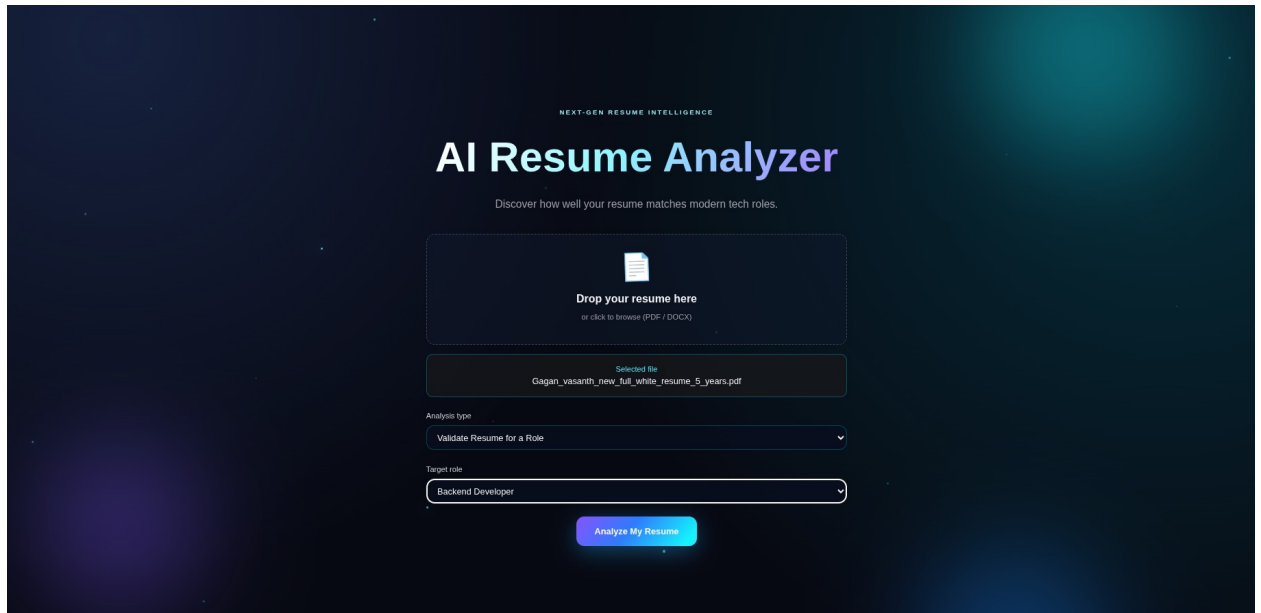
Continued processing state with step completion indicators.

Figure 6: Results Dashboard - Frontend Role Example



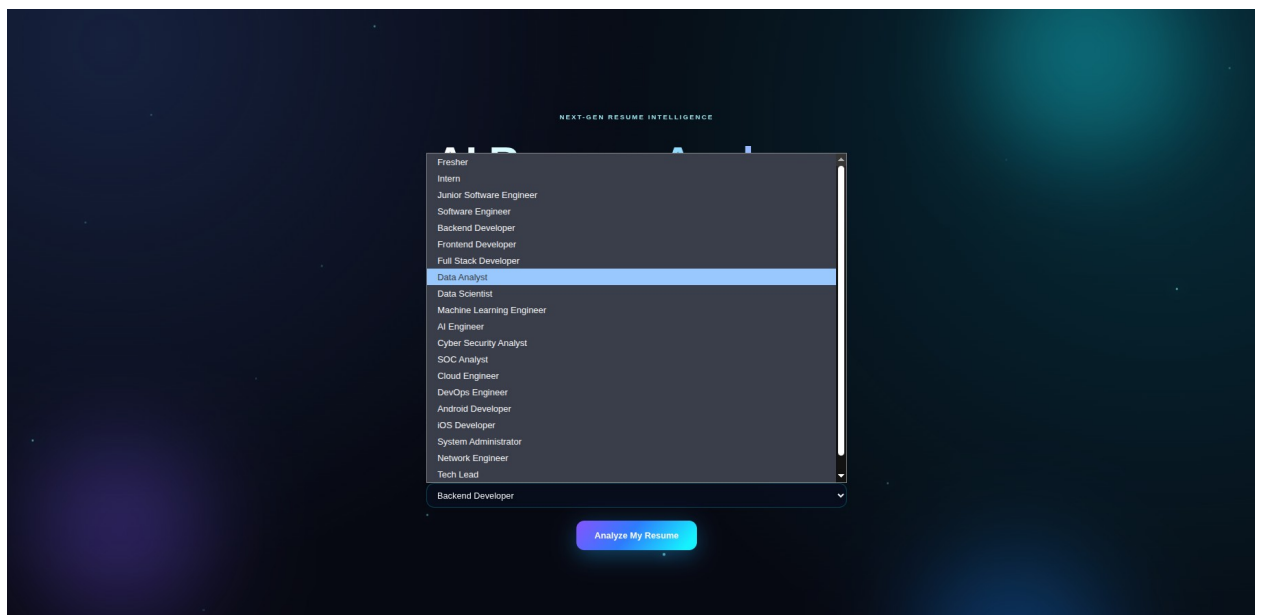
Shows score, matched skills, missing skills, and recommendations.

Figure 7: Results Dashboard - Full Stack Role Example



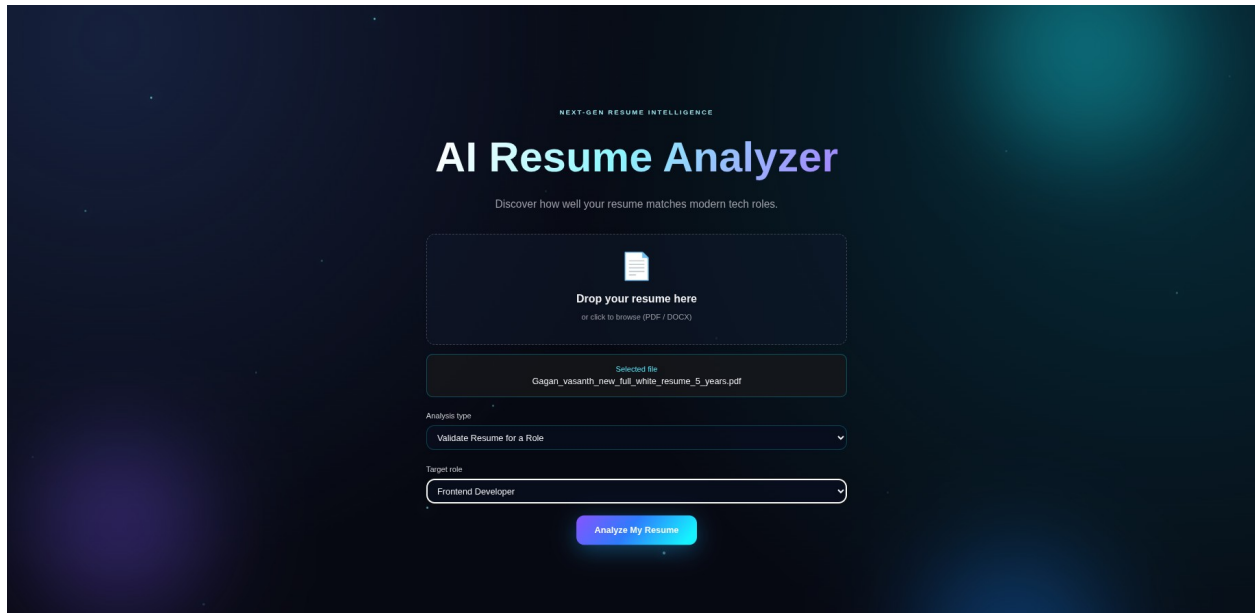
Demonstrates another role outcome with different skill gap distribution.

Figure 8: Dashboard Charts



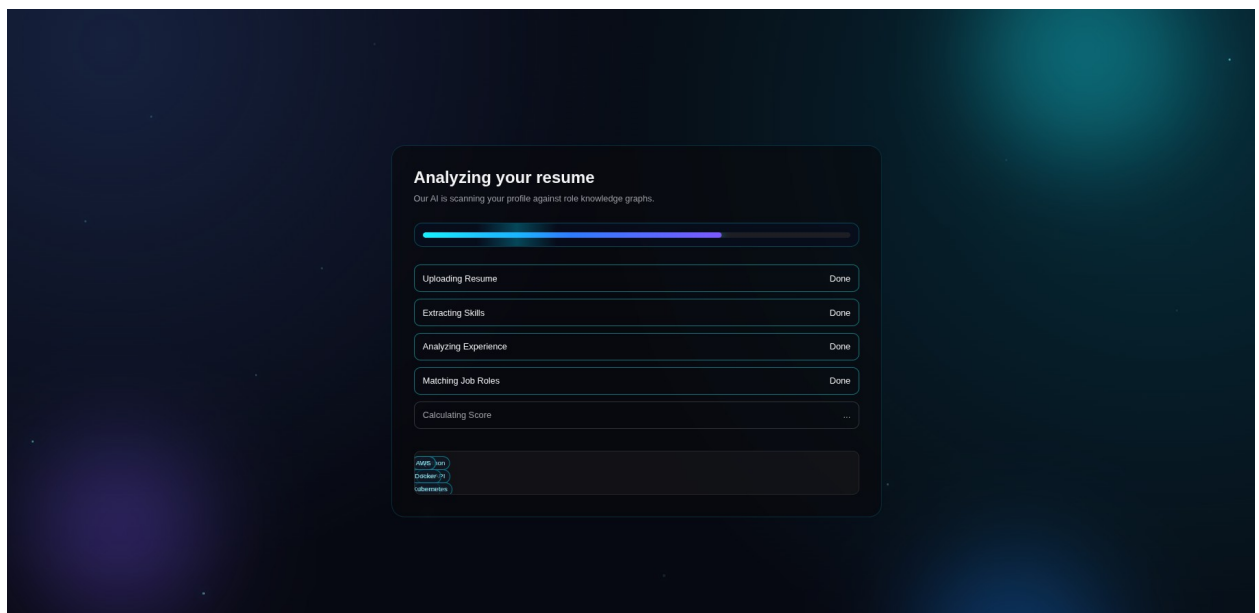
Skill Match Pie, Role Compatibility, and Strength Radar visualizations.

Figure 9: Role Discovery Mode



Find Best Roles workflow where the system predicts top compatible roles.

Figure 10: Invalid Document Validation Warning



Warning state for non-resume uploads blocked by the validation layer.

12. Dashboard and Data Visualization

Skill Match Pie Chart represents matched versus missing skill proportion. Role Compatibility Chart presents top role-fit rankings. Strength Radar Chart provides

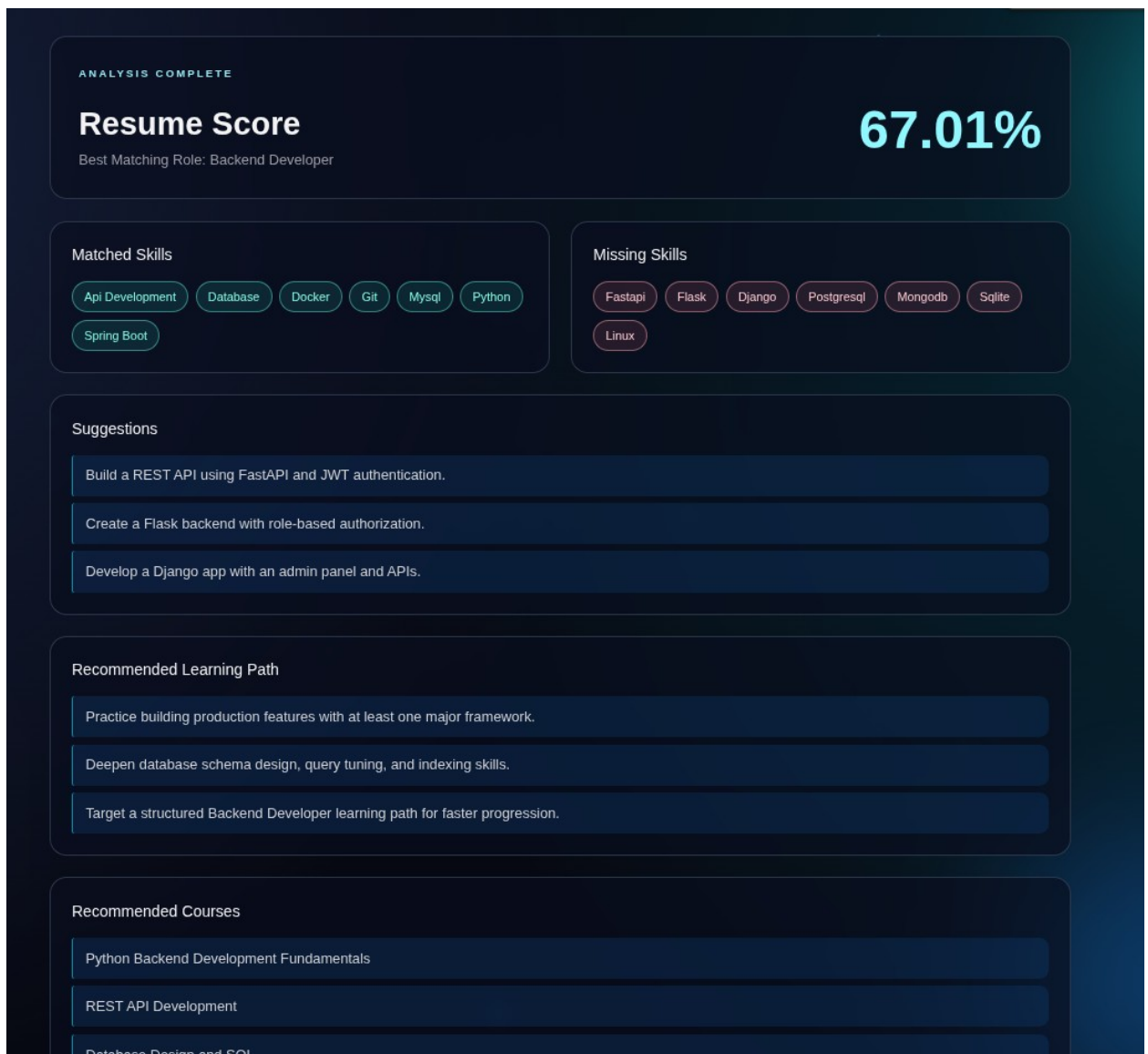
multidimensional readiness visualization across technical depth, core match, tooling, and completeness dimensions.

13. Results and Analysis

The system produces a resume match score, best-matching role, matched skills, missing skills, suggestions, learning path, and recommended courses. These outputs enable users to plan targeted improvement before applying to roles.

Result Screens:

Figure R1: Analysis Result - Pasted image (2)



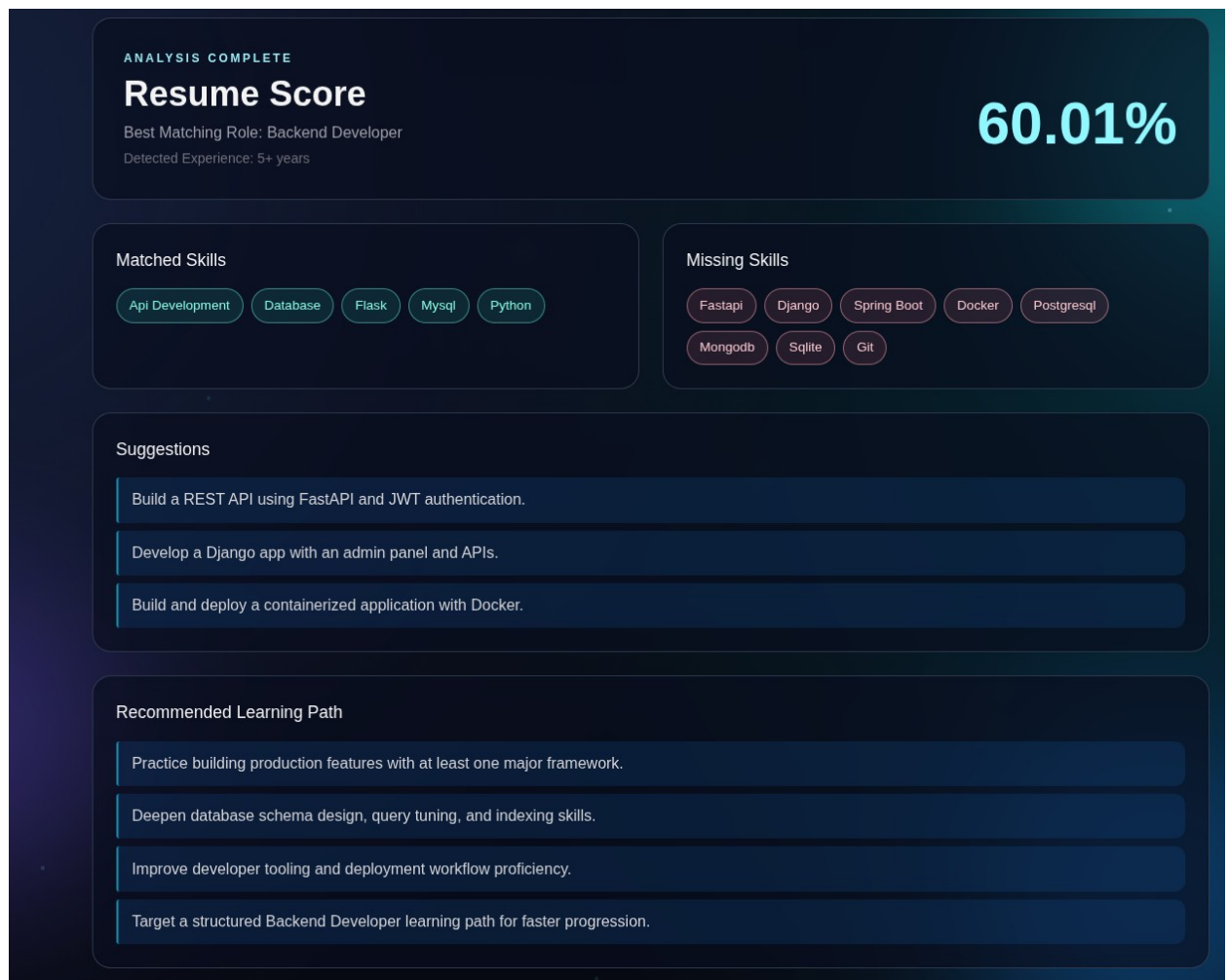
The output summarizes role fit, matched skills, skill gaps, and practical learning recommendations.

Figure R2: Analysis Result - Pasted image (3)

The screenshot shows a web browser window with the URL `localhost:5173`. The browser tabs include "java backend framework", "AI Resume Analyzer", "127.0.0.1:8000", and "Full Video: Ammaadi". The application interface has a dark blue background with the text "NEXT-GEN RESUME INTELLIGENCE" at the top. The main heading is "AI Resume Analyzer" in a large, light blue font, followed by the subtitle "Discover how well your resume matches modern tech roles." Below this is a dashed box containing a document icon and the text "Drop your resume here" or "click to browse (PDF / DOCX)". A file selection box shows "Selected file: Gagan_vasanth_new_full_white_resume_5_years.pdf". There are two dropdown menus: "Analysis type" set to "Validate Resume for a Role" and "Target role" set to "iOS Developer". A blue "Analyze My Resume" button is at the bottom. A red error message at the bottom states: "This document does not appear to be a valid resume. Please upload a recent professional resume that includes sections such as Skills, Education, Work Experience or Projects, and Contact Information. Supported formats: PDF or DOCX."

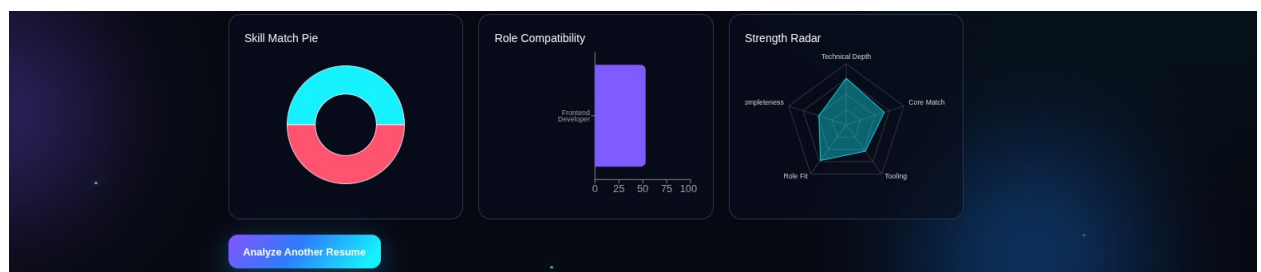
The output summarizes role fit, matched skills, skill gaps, and practical learning recommendations.

Figure R3: Analysis Result - Pasted image (4)



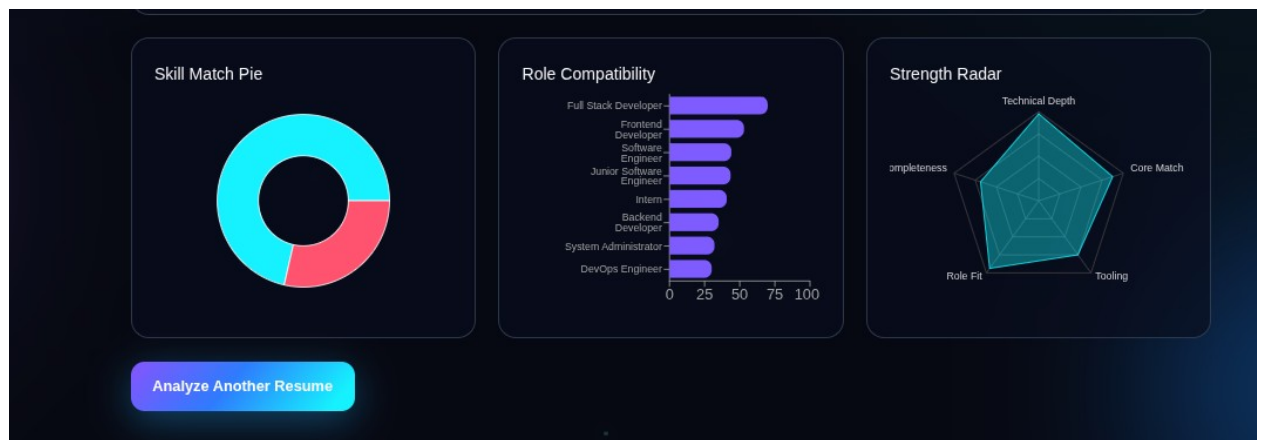
The output summarizes role fit, matched skills, skill gaps, and practical learning recommendations.

Figure R4: Analysis Result - (5)

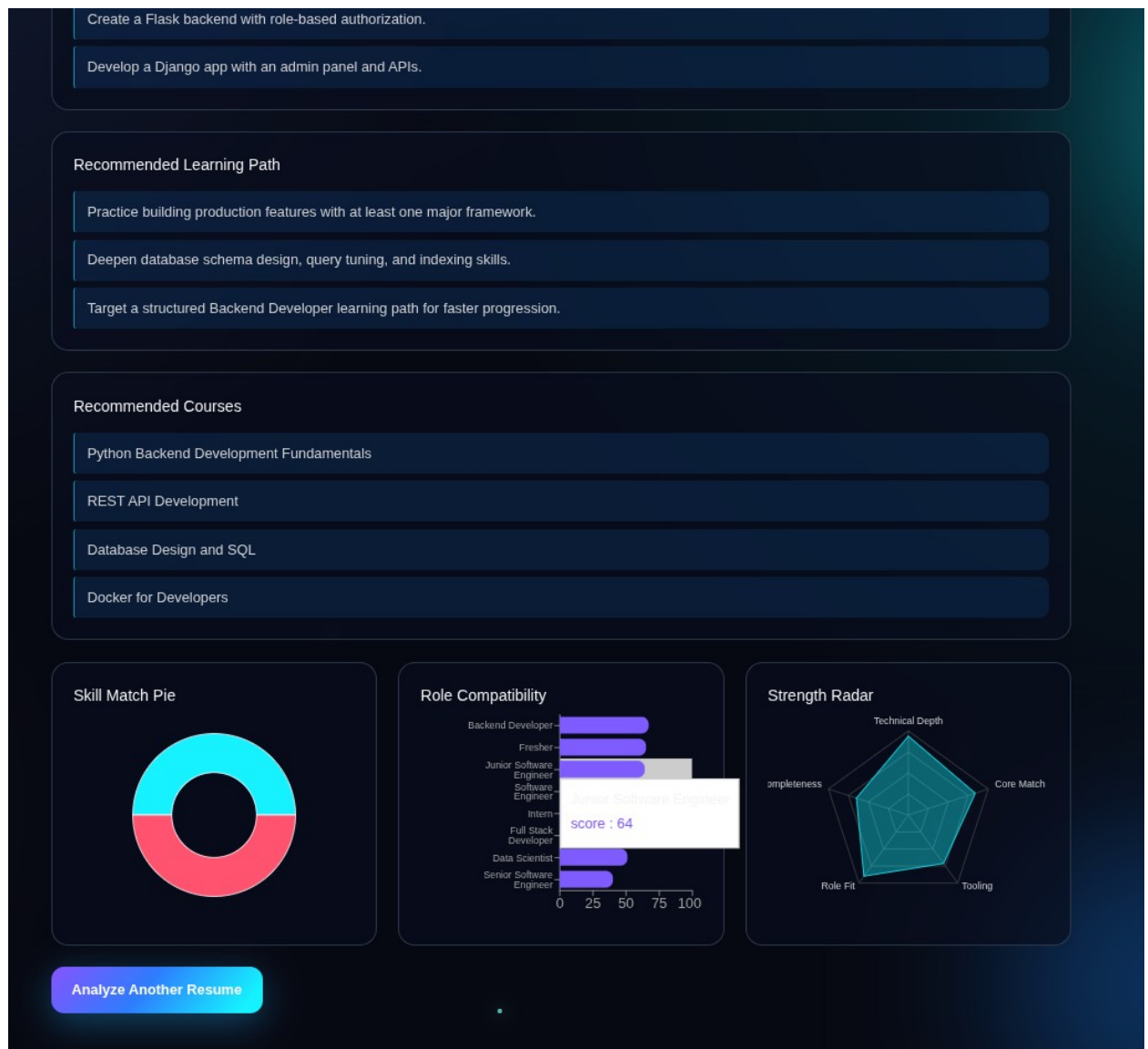


The output summarizes role fit, matched skills, skill gaps, and practical learning recommendations.

Figure R5: Analysis Result - (6)



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13.1 Worst-Case and Edge Case Handling

The platform includes a strict validation layer to prevent misleading analysis from non-resume documents. In worst-case scenarios such as low-content files, academic notes, or unrelated PDFs, the analysis pipeline is intentionally blocked and a clear warning is shown to the user.

14. Advantages of the System

- Improves pre-application career decision quality.

- Provides transparent, explainable skill-gap insights.
- Uses practical AI/NLP for resume understanding.
- Reduces trial-and-error role applications.
- Supports students and professionals with guided upskilling.

15. Future Enhancements

- ATS compatibility scoring.
- AI resume rewriting suggestions.
- LinkedIn and portfolio profile integration.
- Graph-based skill intelligence expansion.
- Personalized learning roadmap and progress tracking.

16. Conclusion

AI Resume Analyzer demonstrates a practical and scalable approach to intelligent resume evaluation. By combining validation, NLP, weighted role matching, and dashboard analytics, it helps candidates understand role readiness and improve employability through clear, data-driven recommendations.